A+ Developer Handoff

Chicago Bare-Metal CME Futures Microstructure Research, Backtesting, and Model-Agnostic Decision Engine

Clean implementation specification for a professional, mathematically governed futures microstructure system using Chicago/Aurora-proximity bare metal, Rithmic live market data and execution, Databento GLBX.MDP3 MBO historical data, HftBacktest queue/latency replay, and controlled Rithmic sim-to-live deployment.

Domain	Specification
Live execution	Chicago/Aurora bare metal -> Rithmic data/execution -> EdgeClear -> Plus500US Financial Services -> CME Globex
Historical research	Databento GLBX.MDP3 MBO event windows -> HftBacktest conversion -> queue/latency replay -> model validation
Target tier	Directly below true CME colocated HFT firms; materially faster and more deterministic than desktop, home internet, webhook, and generic VPS traders
Primary products	MES, MNQ, ES, NQ; ZN/ZB for macro confirmation
Research period	2018-present primary; June 2017-present expansion where data quality is acceptable
Model philosophy	Model-agnostic. Transparent and black-box models are both allowed if validation, risk, and reproducibility requirements pass.

Prepared as a clean developer handoff. This document is an implementation specification, not a trading recommendation.

1. Executive specification

Build a complete CME futures microstructure research and execution stack. The output is not a set of hard-coded trading rules. The output is a validated decision engine that estimates action value under latency, queue position, fill probability, transaction cost, adverse selection, inventory risk, and tail loss.

The system shall be designed around event-driven Level 3 / MBO market states and shall validate all candidate behavior across walk-forward historical regimes before any live deployment.

Mathematical foundation: rigorous probability, stochastic processes, statistics, econometrics/time series, numerical analysis, optimization/control, market microstructure, and robust systems engineering. This foundation aligns with the uploaded quantitative finance framework, which identifies those disciplines as the core backbone for advanced quantitative finance research and production systems.

> A+ standard: every live action must be traceable to a timestamped state, a model version, an expected-value estimate, a risk estimate, execution assumptions, and a pass/fail risk gate.

2. Non-negotiable mathematical invariants

These are structural requirements for mathematical validity. They define the research contract.

- Filtration integrity: at decision time t, features may use only information in F_t, the information set available at or before t.
- Event-time correctness: MBO order-flow is treated as asynchronous marked events, not as evenly spaced candle data.
- Execution realism: predicted edge is evaluated after queue position, latency, fill probability, spread, fees, slippage, and adverse selection.
- Model agnosticism: no model class is preferred in advance. Model approval is empirical and mathematical, not aesthetic.
- Walk-forward evaluation: no random train/test splits across time. Discovery, confirmation, holdout, and recent holdout are strictly separated.
- Action-value output: the model outputs values for actions, not a simple long/short signal.
- No-trade is a valid optimal action.
- Production kill conditions are part of the mathematical system, not operational afterthoughts.

3. Pure mathematical model

The recommended mathematical form is a partially observed marked-point-process stochastic-control model. In production, the components may be estimated by statistical, machine-learning, Bayesian, state-space, or black-box methods. The governing object remains the same: a queue-aware action-value decision system.

3.1 Information set and state

```
F_t = all information available up to time t only.

X_t = (B_t, A_t, Q_t, I_t, Z_t, L_t, E_t)

B_t = bid-side order book state
A_t = ask-side order book state
Q_t = queue / fill state
I_t = current inventory / position
Z_t = latent regime state
L_t = latency state
E_t = event-state label: CPI, NFP, FOMC, prop-flat, cash open, normal
```

The limit order book is represented as queues at price levels:

```
B_t = {(p_i^b, q_i^b)} for i = 1..K
A_t = {(p_i^a, q_i^a)} for i = 1..K
```

3.2 Marked event process

Each MBO action is modeled as a marked event: add, cancel, modify, trade, side, level, size bucket, and event family.

```
N_t^k = counting process for event type k

lambda_k(t) = exp( beta_k' f(X_t) + sum_j integral_0^t phi_kj(t-s) dN_s^j )

Interpretation:
- beta_k' f(X_t): state-dependent event intensity
- phi_kj: self-excitation / cross-excitation across order-flow events
- N_t^k: event-time process, not clock-time bar data
```

3.3 Latent regime posterior

```
Z_t in {normal, event_shock, liquidity_vacuum, stop_cascade,
        prop_flatten, rebuild, chop, trend, spread_stress}

The model estimates P(Z_t = z | F_t), not a hard regime label.
```

3.4 Queue and fill model

```
H_t = queue-ahead volume for the candidate order

tau_fill = inf{u >= t : H_u <= 0}

P_fill(a,h) = P(tau_fill <= t+h | X_t, action a)
```

3.5 Adverse selection

```
For a long fill:
AS_h = (m_{t+h} - p_exec) / tick_size

For a short fill:
AS_h = (p_exec - m_{t+h}) / tick_size

If E[AS_h | fill] < 0, the fill is toxic.
```

3.6 Action-value objective

```
A_t = {no_trade, enter_long, enter_short, add, reduce,
       flatten, cancel, replace, passive_join, marketable_limit}

EV_t(a) = P_fill(a) * E[PnL | fill, a]
          - (1 - P_fill(a)) * C_miss(a)
          - C_fees(a)
          - C_slippage(a)
          - C_latency(a)
          - C_adverse_selection(a)

Q_t(a) = EV_t(a)
         - lambda_tail * ES_alpha(loss | a)
         - lambda_inv * I_t^2
         - lambda_lat * LatencyRisk_t(a)
         - lambda_unc * ModelUncertainty_t(a)

a_t* = argmax_a Q_t(a)
```

3.7 Position sizing

```
size_t = min(
    liquidity_limit,
    margin_limit,
    expected_shortfall_limit,
    daily_loss_limit,
    confidence_scaled_size,
    latency_adjusted_size,
    product_max_size,
    model_max_size
)

Fractional Kelly may be used as one candidate sizing input:
raw_kelly_size = kappa * mu_hat(a) / sigma_hat(a)^2
with kappa in (0,1), capped by all hard risk limits above.
```

4. Model output contract

The live model returns an action-value table. The execution gateway uses this table only after risk approval.

Output	Required meaning
selected_action	Best approved action from the action set
Q_value_by_action	Risk-adjusted value for every candidate action
expected_gross_edge_ticks	Expected gross PnL before costs
expected_cost_ticks	Fees, spread, slippage, missed-fill cost, and latency cost
expected_net_edge_ticks	Gross edge minus all modeled costs
fill_probability	Estimated probability of fill within the target horizon
adverse_selection_ticks	Expected markout after fill
tail_risk	Expected shortfall / p95-p99 loss estimate
latency_sensitivity	Performance impact under latency bands and empirical latency
final_size	Risk-approved contract size
risk_status	PASS, BLOCK, HALT, or FLATTEN

5. Training targets

The model shall not train only on price direction. The target system must include return, execution, adverse-selection, risk, and action-value targets.

Target family	Definition
Forward mid-price return	100 ms, 250 ms, 500 ms, 1 sec, 5 sec, 15 sec, 60 sec
Fill-adjusted realized PnL	PnL after queue, latency, spread, fees, and slippage
Adverse selection	Post-fill markout at 100 ms, 500 ms, 1 sec, 5 sec
Fill probability	By product, order type, queue state, latency band, and event family
Stop-out probability	By entry state, liquidity state, volatility regime, and product
Tail loss	Expected shortfall, worst event-window loss, p95/p99 loss
Action value	Expected value of no trade, passive entry, marketable entry, flatten, cancel, replace

```
Multi-target loss example:
L = w_return * L_return
  + w_fill   * L_fill
  + w_pnl    * L_pnl
  + w_as     * L_adverse_selection
  + w_tail   * L_tail
  + w_cal    * L_calibration
  + Omega(theta)
```

6. Bare-metal and low-latency system requirements

Layer	Requirement
Server	True Chicago/Aurora-proximity bare metal; dedicated CPU/RAM/NVMe; no shared VPS assumption
OS	Ubuntu 22.04 LTS, Ubuntu 24.04 LTS, or Debian stable
CPU	High-clock AMD Ryzen, EPYC, or Intel equivalent; 64 GB RAM minimum, 128 GB preferred
Network	Dedicated public IPv4, stable route to Rithmic Chicago/Aurora, zero packet loss during test
Clock	chrony or PTP-grade clock sync with drift monitoring
Security	SSH key-only, inbound restricted to approved IP/VPN, no public dashboard

6.1 CPU layout

```
Core 0: OS housekeeping, SSH, monitoring
Core 1: Rithmic session / gateway adapter
Core 2: execution gateway, order state, cancel/replace manager
Core 3: risk engine / kill switch
Cores 4-7: strategy workers and feature computation
Remaining cores: async logging, reporting, replay jobs, dashboard
```

6.2 Kernel tuning template

```
isolcpus=2-7 nohz_full=2-7 rcu_nocbs=2-7
processor.max_cstate=0 intel_idle.max_cstate=0 amd_idle.max_cstate=0 cpuidle.off=1

Adjust core ranges to actual CPU topology.
```

6.3 Pre-production latency report

- ping and mtr/traceroute to the Rithmic gateway from the actual server
- packet loss and jitter test across one full CME session
- CPU steal = 0 and no unexplained p99 latency spikes
- disk latency report for async logging path
- clock sync report and drift alert thresholds
- Rithmic sim connectivity and order lifecycle test from the server

7. Live architecture

Component	Responsibilities
Rithmic adapter	Connect to Rithmic, consume market data/depth/MBO where enabled, maintain local book state, handle order entry/cancel/replace/fills/rejects/heartbeat/reconnect
Execution gateway	Receive model actions, run pre-trade risk, construct order intents, manage working orders, track positions, reconcile fills, emergency flatten
Model runtime	Compute state features and action values. Rust/C++/C# preferred for hot path; Python/Numba allowed for research/non-hottest paths
Risk layer	Hard limits, stale-data halt, disconnect halt, position mismatch halt, daily loss halt, max order/cancel rate
Telemetry	Timestamp every market-data, decision, order, cancel, fill, reject, halt, and reconnect event

Hot-path communication shall use lock-free queues, shared memory ring buffers, Aeron, or direct in-process modules. Redis/Valkey may be used for dashboards and asynchronous telemetry, not as a live signal-to-order dependency.

8. Historical data plan

Historical research uses Databento GLBX.MDP3 MBO data and HftBacktest replay. Data requests are controlled by event windows, cost estimates, and manifests.

Field	Requirement
Dataset	GLBX.MDP3
Schema	mbo
Primary symbols	MES.v.0, MNQ.v.0, ES.v.0, NQ.v.0
Macro confirmation	ZN.v.0, ZB.v.0
Optional expansion	RTY.v.0, M2K.v.0, CL.v.0, GC.v.0
Primary study	2018-present
Expansion study	June 2017-present where data quality is acceptable

8.1 Contract roll handling

- Store requested continuous symbol, resolved raw contract, Databento instrument ID, roll date, and request hash for every request.
- Verify no duplicate or overlapping roll windows.
- Separate roll-period effects from event-window alpha results.
- All artifacts must be reproducible from events.csv and manifest.parquet.

9. Event-window specification

All event anchors are stored in original timezone and converted to UTC for requests. Event windows are parameterized in events.csv.

Event family	Anchor	Tight	Main	Context	Symbols
CPI	08:30 ET	08:29:30-08:35 ET	08:25-08:45 ET	08:00-09:30 ET	MES/MNQ/ES/NQ/ZN/ZB
NFP / Employment Situation	08:30 ET	08:29:30-08:35 ET	08:25-08:45 ET	08:00-09:30 ET	MES/MNQ/ES/NQ/ZN/ZB
FOMC statement	14:00 ET	13:59:30-14:05 ET	13:55-14:15 ET	13:30-15:00 ET	MES/MNQ/ES/NQ/ZN/ZB
FOMC press conference	14:30 ET	14:29:30-14:35 ET	14:25-14:45 ET	14:25-15:30 ET	MES/MNQ/ES/NQ/ZN/ZB
Cash equity open	09:30 ET	09:29:30-09:35 ET	09:25-09:45 ET	09:15-10:00 ET	MES/MNQ/ES/NQ
Topstep-style forced flatten	15:10 CT	15:05-15:12 CT	14:45-15:20 CT	14:30-15:30 CT	MES/MNQ/ES/NQ
TPT-style late flatten	16:55 ET / 15:55 CT	15:45-16:00 CT	15:30-16:00 CT	15:00-16:00 CT	MES/MNQ/ES/NQ
MFFU news restriction	event time	T-2 to T+2 min	T-5 to T+15 min	T-30 to T+30 min	MES/MNQ/ES/NQ/ZN/ZB
Friday de-risking	Friday close	15:30-16:00 CT	14:30-16:00 CT	13:30-16:00 CT	MES/MNQ/ES/NQ

10. Events registry

events.csv is the authoritative event registry. Every rule-based or prop-firm window must carry a source, effective date, and parse version.

```
event_id,event_type,release_date,release_time,timezone,window_name,
start_offset_seconds,end_offset_seconds,symbols,priority,source,source_url,effective_date,notes

CPI_2024_09_11_TIGHT,CPI,2024-09-11,08:30:00,America/New_York,TIGHT,
-30,300,"MES.v.0,MNQ.v.0,ES.v.0,NQ.v.0,ZN.v.0,ZB.v.0",1,BLS,<url>,2024-01-01,"CPI tight window"

TOPSTEP_2024_09_18_MAIN,PROP_FLATTEN_TOPSTEP,2024-09-18,15:10:00,America/Chicago,MAIN,
-1500,600,"MES.v.0,MNQ.v.0,ES.v.0,NQ.v.0",1,TOPSTEP,<url>,2024-01-01,"forced-flat window"
```

11. Databento credit mathematics

Before every download, the system calls metadata.get_cost and records the result in a manifest. The initial $125 credit is treated as a budgeted research asset.

```
cost_i = metadata.get_cost(
    dataset = "GLBX.MDP3",
    schema  = "mbo",
    symbols = symbols_i,
    start   = start_i,
    end     = end_i
)

total_used = sum(cost_i)
credit_remaining = 125.00 - total_used

duration_seconds_i = end_i - start_i
symbol_count_i = len(symbols_i)

cost_per_symbol_second_i = cost_i / (symbol_count_i * duration_seconds_i)
cost_per_symbol_minute_i = cost_i / (symbol_count_i * duration_seconds_i / 60)

forecast_cost = median(cost_per_symbol_minute by event_type)
                * planned_symbol_count
                * planned_total_minutes
```

Budget rule	Value
Initial credit	$125.00
Operating cap	$112.50
Reserve	$12.50
Single request soft limit	$5.00
Single request hard limit	$10.00 unless explicitly approved

12. Data pull sequence

Phase	Years	Events	Symbols	Output
0 - Cost calibration	2023-2025	3 CPI, 3 NFP, 3 FOMC statements, 3 FOMC press conferences, 10 prop windows, 10 cash opens	MES/MNQ/ES/NQ	Cost per symbol-minute by event, symbol, and regime
1 - Core research	2018-present	All CPI, NFP, FOMC, prop windows, Friday late-day, monthly sampled cash opens	MES/MNQ/ES/NQ	Primary event-window training data
2 - Macro confirmation	2018-present	CPI, NFP, FOMC statement, FOMC press conference	Add ZN/ZB	Rates-equity impulse tests
3 - Full clean expansion	June 2017-present	All event families where data quality is acceptable	Primary + macro	Maximum clean MBO regime coverage
4 - Final holdout	2025-present + latest 60 CME days	No model design use	MES/MNQ first	Final approval only

13. HftBacktest pipeline

The backtest layer converts Databento MBO files into HftBacktest-compatible arrays, then replays the event windows under queue, latency, fee, fill, and adverse-selection assumptions.

```
Input:  raw .dbn.zst / .mbo.dbn.zst
Output: .npz arrays for HftBacktest

Required conversion metadata:
- event_id, event_type, symbol, resolved contract, instrument ID
- input_file, output_file, start_utc, end_utc
- row_count, conversion_status, data_quality_status

Queue models:
- LogProbQueueModel2
- SquareProbQueueModel

Exchange model:
- NoPartialFillExchange

Latency models:
- fixed bands: 0.5 ms, 1 ms, 2 ms, 5 ms, 10 ms
- empirical latency model after Rithmic samples exist
```

14. Feature set

All features must be computable using only data available up to timestamp t.

Feature group	Features
Order-flow	aggressor_volume_imbalance, buy_aggressor_volume, sell_aggressor_volume, cancel_to_add_ratio, cancel_burst_score
Depth/queue	top_1_depth, top_3_depth, top_5_depth, top_10_depth, queue_depletion_rate, refill_speed, refill_ratio
Liquidity stress	spread_stress, book_slope, book_slope_change, liquidity_vacuum_score, near_touch_cancel_pressure
Absorption/reload	absorption_score, iceberg_reload_score, passive_trap_score, adverse_selection_after_fill
Lead-lag	lead_lag_ms, micro_vs_mini_lag, rates_equity_impulse_score
Prop/event	forced_flatten_score, cutoff_pressure_score, prop_reentry_score, news_restriction_flatten_score
State	realized_latency_state, volatility_state, liquidity_state, regime_state

15. Candidate L3 / microstructure hypothesis families

These are hypothesis families and feature families. The final model may use them directly, combine them nonlinearly, or discover alternatives. Each family produces a research card, not a standalone production bot.

#	Hypothesis family	MBO signature / behavior
1	Second-wave continuation	Initial sweep followed by continued same-side aggressor flow
2	Stop-run exhaustion fade	Sweep through obvious level, stop spike, aggressor deceleration, opposite-side refill
3	Liquidity vacuum continuation	Top 3-10 levels vanish and fail to refill
4	Depth-refill imbalance	One side rebuilds faster and more persistently after shock

#	Hypothesis family	MBO signature / behavior
5	Spread blowout/recompression	Spread widens violently, then compresses with improving depth
6	Aggressor deceleration fade	Market orders continue but generate declining price impact
7	Forced liquidation cascade	One-way aggressive flow, widening spread, weak refill
8	False breakout trap	Break through level with weak MBO follow-through
9	Cancel storm before move	Large near-touch cancels before price break
10	Queue depletion trigger	Best bid/ask drains faster than refill
11	Book slope collapse	Depth disappears across top levels
12	Absorption fade	Aggressors hit same level; price does not move
13	Iceberg/reload detection	Displayed level trades and replenishes repeatedly
14	Liquidity defense break	Defended level stops reloading
15	One-sided add/cancel imbalance	Adds stack one side while cancels remove opposite side
16	ES -> MES lead-lag	ES order-flow pressure leads MES reaction
17	NQ -> MNQ lead-lag	NQ order-flow pressure leads MNQ reaction
18	ES/NQ divergence snapback	Short-term correlation gap opens and reverts
19	ZN/ZB -> ES/NQ macro impulse	Rates react first around CPI/FOMC; equities respond after
20	Micro contract retail lag	Main contract moves first; micro reacts slower/noisier
21	Round-number stop sweep	Stop flow near obvious handles, prior high/low, premarket levels
22	Prior high/low breakout trap	Breakout with weak depth/refill/aggressor confirmation
23	Opening candle chase	09:30 ET impulse followed by late chase or reversal
24	VWAP defense/break	Repeated reload near VWAP followed by hold or failure
25	DOM illusion trap	Large displayed size cancels before price reaches it
26	Late candle entry fade	Visible candle expansion with weakening flow
27	Stop-loss cascade continuation	Stops convert into market orders and book stays thin
28	Panic market-order spread tax	Forced orders cross widened spread during stress
29	End-of-day forced flatten flow	Directional exit pressure into known flat windows
30	Cutoff panic exits	Urgent flattening in final minutes before cutoff
31	No-overnight inventory squeeze	Late-day one-way reduction in directional exposure
32	Daily loss-limit defense	Sudden flattening after adverse intraday move
33	Trailing drawdown pressure	Panic exit after reversal threatens account state
34	Profit-lock behavior	Trend-day winners close into payout/cutoff protection
35	Max-contract crowding in micros	MES/MNQ show noisier forced flow than ES/NQ
36	Prop reset/reopen window	Re-entry behavior after allowed session resumes
37	Friday/weekend de-risking	Friday final-hour exposure reduction
38	Economic-event restriction flattening	Pre-news flattening and post-news re-entry
39	Quote pull before volatility	Depth vanishes before scheduled high-impact events
40	Re-quote race after shock	Depth returns unevenly after event shock
41	Thin-book continuation	Market makers remain wide after first move
42	Passive trap fill	Passive fills followed by immediate adverse selection
43	Rebate trap avoidance	Maker-style fills lose more to adverse selection than spread/rebate benefit
44	Spread regime change	Spread remains elevated longer than baseline after shock

16. Validation standard

Split	Years / rule
Discovery	2018-2020
Confirmation	2021-2022
Holdout	2023-2024

Split	Years / rule
Recent holdout	2025-present
Live/sim shadow	latest 60 CME trading days

16.1 Required tests

- walk-forward testing only
- event-family separation and product separation
- latency-band testing at 0.5 ms, 1 ms, 2 ms, 5 ms, and 10 ms
- fee, spread, slippage, missed-fill, queue, and adverse-selection modeling
- multiple-testing control and deflated Sharpe or equivalent penalty
- confidence calibration and probability calibration
- feature leakage audit and timestamp audit
- p95/p99 latency sensitivity and degradation analysis
- live/sim disagreement report before live deployment

16.2 Required metrics

Category	Metrics
Performance	net PnL, expectancy per trade, Sharpe, deflated Sharpe, Sortino, profit factor
Risk	max drawdown, expected shortfall, p95/p99 loss, tail loss
Execution	fill rate, missed-fill rate, cancel rate, reject rate, adverse selection at 100 ms / 500 ms / 1 sec / 5 sec
Robustness	event-family performance, product performance, holdout performance, latency sensitivity
Capacity	turnover, capacity estimate, liquidity impact, order-rate utilization

17. Model artifact and research card

Every approved model and every rejected model produces a permanent artifact. This prevents silent overfitting and allows later audit.

```
model_id:
model_version:
model_class:
alpha_family_or_discovered_behavior:
training_period:
validation_period:
holdout_period:
symbols:
event_windows:
features:
targets:
latency_assumptions:
queue_model:
fill_model:
cost_model:
risk_model:
decision_rule:
position_sizing_rule:
risk_constraints:
known_failure_modes:
approval_status:
```

Research cards shall include years tested, events tested, features used, model class, target labels, latency bands, queue model, gross/net PnL, expectancy, win rate, fill quality, adverse selection, tail loss, expected shortfall, pass/fail status, and sim/live recommendation.

18. Production failure states

Production safety states are part of the mathematical system. They are not optional operator conveniences.

Failure state	Required action
stale market data	halt entries; flatten or hold according to risk state
Rithmic disconnect	halt entries; reconcile position; alert operator
clock drift	halt live trading until clock integrity restored
latency spike	block new entries if above empirical p99 threshold
missing fill reconciliation	halt and reconcile broker/order state
order reject escalation	block affected product/model and alert
position mismatch	halt and flatten if mismatch cannot be resolved
daily loss limit	flatten and halt

Each halt event must include reason, timestamp, open orders, current position, attempted action, operator alert, and recovery requirement.

19. Deliverables for developer quote

Package	Deliverables
1. Bare-metal server build	Linux install, CPU isolation, process pinning, IRQ tuning, clock sync, security hardening, Rithmic connectivity test, latency report
2. Rithmic gateway	market data listener, order entry, cancel/replace, fill/reject handling, reconnect, heartbeat, position state, emergency flatten, latency logger
3. Databento research data system	event calendar builder, cost estimator, budget-controlled downloader, manifest.json, manifest.parquet, cost-per-symbol-minute report, duplicate prevention, UTC conversion, contract roll metadata
4. HftBacktest pipeline	Databento conversion, queue models, latency bands, empirical latency support, replay runner, model plug-in interface, fee/slippage model
5. Feature and hypothesis research	core MBO features, all 44 hypothesis families, event-window reports, research cards, ranking board, latency sensitivity board
6. Model training and decision engine	model-agnostic training, target construction, leakage controls, walk-forward validation, model registry, action-value output, sim-ready inference runtime
7. Risk and production readiness	max position/loss/order-rate/cancel-rate controls, kill switch, stale-data halt, disconnect halt, position mismatch handling, sim-to-live checklist
8. Reporting dashboard	latency, PnL, fill quality, adverse selection, event comparison, symbol comparison, model ranking, deploy/sim/reject board

20. Acceptance criteria

- Bare-metal server configured and latency-tested from the actual production host.
- Rithmic sim/live connectivity works from the server.
- Databento downloader works with cost controls and produces reproducible manifests.
- All event windows are reproducible from events.csv.
- HftBacktest conversion works from raw DBN/ZST to replayable arrays.
- Contract roll handling is reproducible and audited.
- All required MBO features are computed without leakage.
- All 44 hypothesis families are represented as testable research modules.
- Model training supports transparent and black-box models.
- Latency bands are tested at 0.5, 1, 2, 5, and 10 ms.
- Empirical Rithmic latency model is produced once samples exist.
- Models are ranked by net expectancy, fill quality, latency sensitivity, adverse selection, and tail risk.
- MES/MNQ sim testing is completed before ES/NQ live consideration.
- Risk layer can block, flatten, and halt trading.
- Final report shows Databento credit used, remaining credit, and forecast expansion cost.
- Final model approval requires holdout success and Rithmic sim-shadow confirmation.

21. Source anchors

The implementation shall cite exact source URLs in the repository documentation and store source effective dates for all event windows and prop-firm rules.

Source	Use in specification
Databento GLBX.MDP3 documentation	Dataset, schema, continuous symbology, historical API, get_cost, DBN/ZST handling
HftBacktest documentation	Databento converter, queue models, fill model limitations, replay assumptions
BLS CPI release schedule	CPI release dates and 08:30 ET anchors
BLS Employment Situation release schedule	NFP/Employment Situation release dates and 08:30 ET anchors
Federal Reserve FOMC calendar	FOMC dates, statement and press conference anchors
Topstep help documentation	Topstep-style forced-flat window source
MyFundedFutures help documentation	Tier 1 news restriction / re-entry window source
Take Profit Trader help documentation	TPT-style late flatten window source
Uploaded quantitative finance framework	Mathematical backbone: probability, statistics, time series, numerical methods, optimization/control, finance, and systems

If a prop-firm rule source changes, the event registry must preserve the prior effective date and create a new source-versioned rule. This keeps historical experiments reproducible and prevents stale rule assumptions from contaminating model validation.

22. Quote request

Provide a quote with fixed-price and hourly estimates by package, timeline by package, required credentials, required server access, required Rithmic access, required Databento access, external dependencies, risks/blockers, testing plan, production-readiness checklist, and maintenance estimate after delivery.

The quote should assume a professional-grade Chicago futures microstructure system using bare-metal infrastructure, Rithmic live data/execution, Databento historical MBO data, HftBacktest queue/latency replay, prop-firm flow modeling, model-agnostic decision logic, 2018-present event-window validation, empirical latency modeling, and controlled sim-to-live deployment.